

Analysis of Delaware Real Estate Data Set

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MATH 4339 - 15445

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Data Set Introduction

The data set used in the this project is a subset of the USA Real Estate data set posted on Kaggle by the author, Mr. Sakib, who extracted the data from realtor.com, which is the second most visited real estate website in the U.S. with over 100 million active users. The original data set included an excess of 20K listings spanning 18 states. To make it easier to perform analysis, the subset was created by focusing solely on the state of Delaware with around 2K listings and 6 columns for variables. Through performing analysis on this subset, the goal is to primarily find answers to two important questions.

- **Question 01:** Does the size of the house(sq-ft, # of beds, # of baths) vary significantly based on ZIP code?
- **Question 02:** Does the price of the house vary significantly based on size of the house (sq-ft, # of beds, # of baths) ?

Through the applications of multivariate data analysis techniques learned through the course of this semester such as MANOVA, PCA, and Multiple Linear Regression we hope to find answers to these questions and to expand our understanding of Multivariate Data Analysis.

Also, since ZIP code is an ubiquitous concept used daily, we wanted to have a closer look at its origin and reasons behind the popularity. Commonly consisting of 5 digits, the U.S. ZIP codes, or **Zone Improvement Code Plan code**, was first implemented in 1963, thanks to Mr. Robert Moon, who developed the system after a 20-year development process. Prior to its implementation, USPS used a less structured system that was only applicable to cities. The current system evolved out of necessity due to a rapid swell in the population, as well as a boom in commerce following the first and second World War eras.

There are three main parts of the 5-digit ZIP Code — the national area, the region or city, and the delivery area. The United States Postal Service (USPS) has segmented the country into 10 ZIP Code areas. Starting in the northeast, they are number 0-9. The addition of the 4 digits in the 1980s called the ZIP+4 system, allowed senders to indicate an even more precise location, such as particular block or apartment building and even which side of the street. So that is a quick take on ZIP codes. Now, let's bring the focus back to the data set in question.

Data set Analysis

- **Assessing and cleansing the data set**

code:

```
realtor = read.csv(file.choose())
cleanRealtor = na.omit(realtor)

state_to_filter = "Delaware"
subset_data_state <- cleanRealtor[cleanRealtor$state == state_to_filter, ]

numeric_data <- subset_data_state[, sapply(subset_data_state, is.numeric)]
```

Will be **grouping the ZIP codes** and analyzing the data for **change of variance among** the 4 response variables (“number of baths”, “number of beds”, “size of the lot”, and “house size”) using MANOVA.

- **Find the unique ZIP codes**

code:

```
unique_values <- unique(numeric_data $ zip_code)
sorted <- sort(unique_values) #used to sort in ascending order
print(sorted)
```

```
## [1] 19701 19702 19703 19706 19707 19709 19711 19713 19720 19730 19734 19801
## [13] 19802 19803 19804 19805 19806 19807 19808 19809 19810 19901 19977
```

- **Sort the 23 ZIP codes into 3 groups** (8 in first two 7 in the last one) in order to perform **MANOVA**

```
Group01: 19701 19702 19703 19706 19707 19709 19711 19713
Group02: 19720 19730 19734 19801 19802 19803 19804 19805
Group03: 19806 19807 19808 19809 19810 19901 19977
```

- **Sorting the entire data set** ascending order in terms of ZIP_code

code:

```
sorted_RE <- numeric_data[order(numeric_data $ zip_code), ]
View(sorted_RE)
```

- **Create 3 groups based** on ranges

```
sorted_RE$groups <- cut(sorted_RE$zip_code, breaks = c( 0,19720, 19806, 99999999), labels = c("Group 1"
attach(sorted_RE)
```

- **Hypothesis of the Test**

Null: The three group mean vectors are equal to each other.

Alt: At least one of the group mean vector is different.

- **Assumptions of the Test**

- Multivariate Normal - Hypothesis test
- No sub populations within the 3 species.
- Common Variance - Covariance Matrix - BoxM test
- Independent samples

Since the data set is quite large, we can assume MVN based on the Central Limit Theorem.

Next, use the **BoxM test** to check Homogeneity of Covariance Matrix.

```
code:
library(biotools)
boxM(sorted_RE[, c(1, 2, 3, 5)], sorted_RE[,7])

## Warning: package 'biotools' was built under R version 4.1.3

## Loading required package: MASS

## ---
## biotools version 4.2

##
## Box's M-test for Homogeneity of Covariance Matrices
##
## data: sorted_RE[, c(1, 2, 3, 5)]
## Chi-Sq (approx.) = 2791.5, df = 20, p-value < 2.2e-16
```

In the output, it is shown that the **p-value is 2.2e-16**. This means that there is **strong evidence** that the **variance-covariance matrix is not the same among all three groups**.

- Perform the MANOVA. Give important details

code:

```
RE_data.1 = manova(cbind(bed, bath, acre_lot, house_size) ~ groups, data = sorted_RE)
summary(RE_data.1, test="Wilks")
```

```
##              Df    Wilks approx F num Df den Df    Pr(>F)
## groups        2 0.83621   39.787      8   3402 < 2.2e-16 ***
## Residuals 1704
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value is listed as 2.2e-16 in the summary, which indicates that there is strong evidence against the NULL hypothesis of the MANOVA. At least, one of the ZIP code based groups have a difference in number of beds, number of baths, acre_lot or house_size.

Test-statistic: 39.787

- Bonferroni corrected ANOVAs to assess the significance of individual variables

code:

```
RE_data.lm1 = lm(bed ~ groups , data = sorted_RE)
anova(RE_data.lm1)
```

```
RE_data.lm2 = lm(bath ~ groups , data = sorted_RE)
anova(RE_data.lm2)
```

```
RE_data.lm3 = lm(acre_lot ~ groups , data = sorted_RE)
anova(RE_data.lm3)
```

```
RE_data.lm4 = lm(house_size ~ groups , data = sorted_RE)
anova(RE_data.lm4)
```

****Create box plots**

```
par(mfrow=c(1,2))
boxplot(bed ~ groups,data = sorted_RE)
boxplot(bath ~ groups,data = sorted_RE)
boxplot(acre_lot ~ groups,data = sorted_RE)
boxplot(house_size ~ groups,data = sorted_RE)
```

****Create a profile plot**

```
library(profileR)
library(ggplot2)
group_1 <- subset(sorted_RE, groups == "Group 1")
group_2 <- subset(sorted_RE, groups == "Group 2")
group_3 <- subset(sorted_RE, groups == "Group 3")
selected_group1 = group_1[, c("bed", "bath", "acre_lot")]
selected_group2 = group_2[, c("bed", "bath", "acre_lot")]
selected_group3 = group_3[, c("bed", "bath", "acre_lot")]
```

```

m = rbind(colMeans(selected_group1), colMeans(selected_group2), colMeans(selected_group3))
rownames(m, do.NULL = F)
rownames(m) = c("Group 1", "Group 2", "Group 3")
profileplot(m, rownames(m), standardize = F)

selected_group1 = group_1[, c("bed", "bath", "acre_lot", "house_size")]
selected_group2 = group_2[, c("bed", "bath", "acre_lot", "house_size")]
selected_group3 = group_3[, c("bed", "bath", "acre_lot", "house_size")]
n = rbind(colMeans(selected_group1), colMeans(selected_group2), colMeans(selected_group3))
profileplot(n, rownames(m), standardize = F)

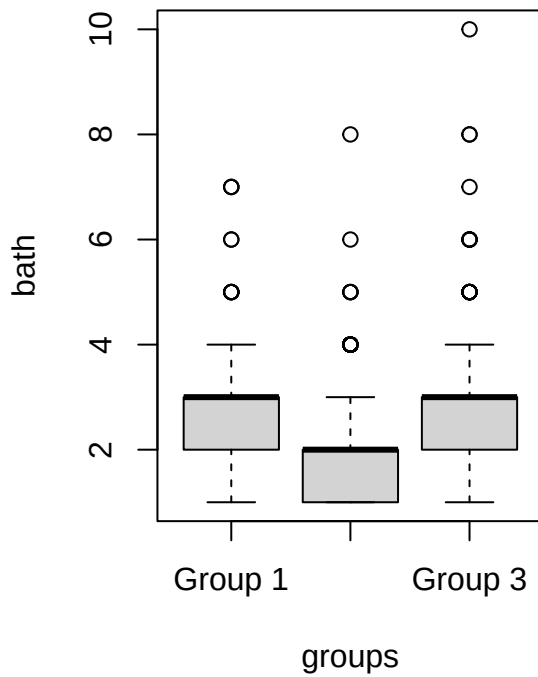
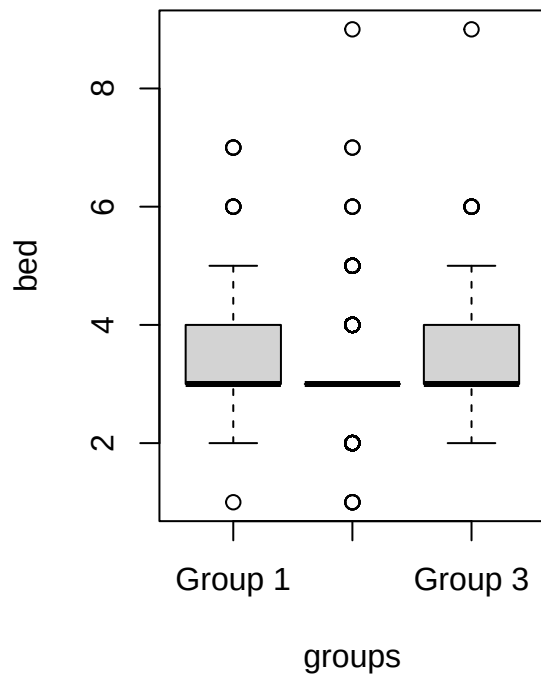
## Analysis of Variance Table
##
## Response: bed
##           Df Sum Sq Mean Sq F value    Pr(>F)
## groups      2   35.29  17.6466   25.444 1.294e-11 ***
## Residuals 1704 1181.82   0.6936
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Analysis of Variance Table
##
## Response: bath
##           Df Sum Sq Mean Sq F value    Pr(>F)
## groups      2  245.0  122.500   132.07 < 2.2e-16 ***
## Residuals 1704 1580.6   0.928
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Analysis of Variance Table
##
## Response: acre_lot
##           Df Sum Sq Mean Sq F value    Pr(>F)
## groups      2   3094  1547.1   1.3305 0.2646
## Residuals 1704 1981293  1162.7

## Analysis of Variance Table
##
## Response: house_size
##           Df Sum Sq Mean Sq F value    Pr(>F)
## groups      2 271284326 135642163  122.51 < 2.2e-16 ***
## Residuals 1704 1886664680  1107198
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```



```
## Warning: package 'profileR' was built under R version 4.1.3

## Loading required package: ggplot2

## Loading required package: RColorBrewer

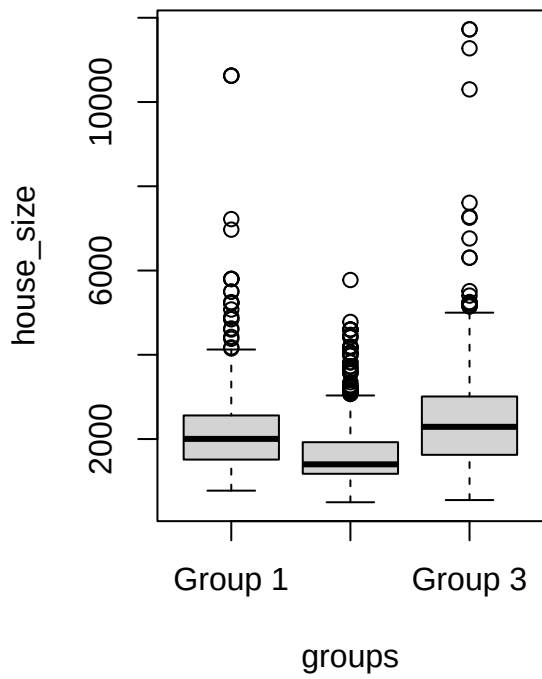
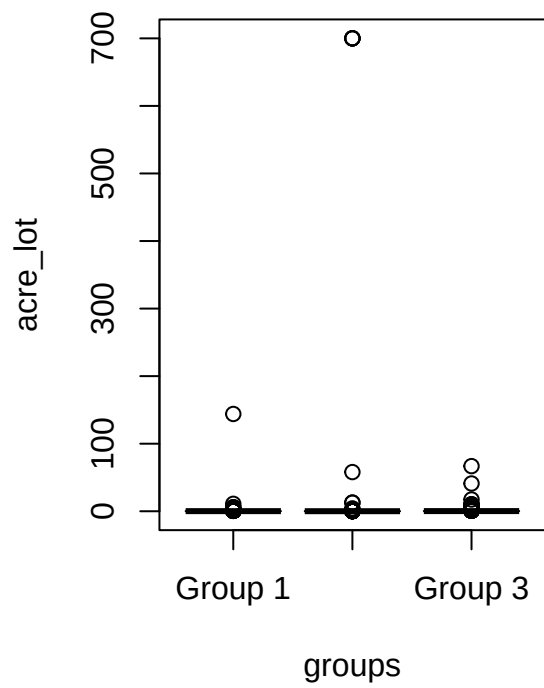
## Warning: package 'RColorBrewer' was built under R version 4.1.3

## Loading required package: reshape

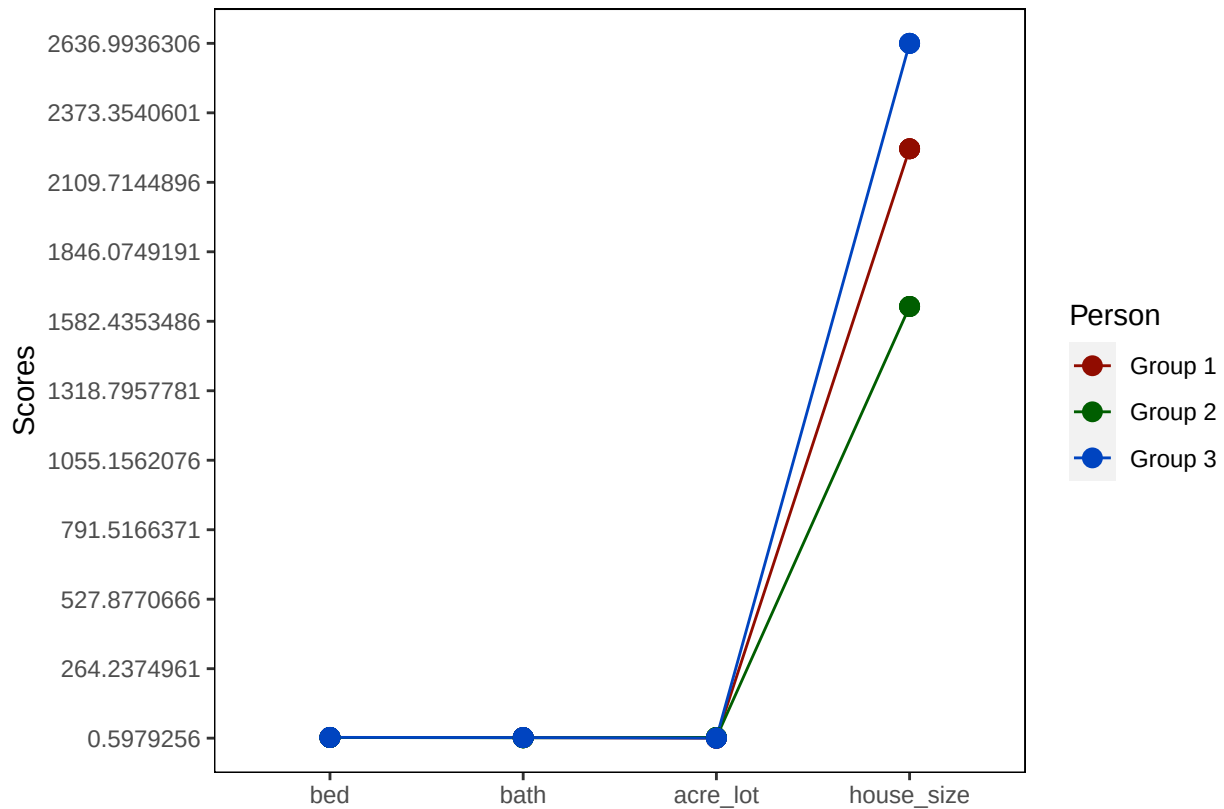
## Warning: package 'reshape' was built under R version 4.1.3

## Loading required package: lavaan

## This is lavaan 0.6-16
## lavaan is FREE software! Please report any bugs.
```



```
## [1] "row1" "row2" "row3"
```



After reviewing the ANOVA tables for each of the variables, `acre_lot` is the only one with a p-value greater than $(0.05/3 = 0.016)$. This means that there is difference in means for each variable except for `acre_lot` among the 3 ZIP code groups.

The size of the lot could be similar for the 3 group, however the size of the house built on the lot is different due to income gap between different areas.

Data Set Summary

- Based on the Box plots, Group1 and Group 3 ZIP codes appears to have consistently larger values and greater variance than Group2.
- These zip codes could possibly be inhabited by families with higher incomes. While Group2 area probably could be inhabited by lower income families. Higher income usually results in larger houses.
- Also when observing the profile plot, it is evident that the largest houses in the state are located in areas covered by Group3 ZIP codes.
- Insights like these could be used by:
 - Retail businesses to find the most suitable spot to start their next branch.
 - Banks to decide on interest rates for consumers.

- – Non-profits to decide on the parts of the state that needs the most attention.
- – Law enforcement agencies to understand and mitigate crime rates within the state
- – AND SO MUCH MORE APPLICATIONS.

References

- Data set - [kaggle.com](https://www.kaggle.com)
- Zip code info - [loqate.com](https://www.loqate.com) , [smarty.com](https://www.smarty.com)